

End Sem, Revision Session-1
Mathematics for Data Science - 1

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$$A = \{1, 2\}$$

$$R = \{(1, 2), (2, 2)\}$$

$$\rightarrow \begin{matrix} (1, 2) & (2, 2) \\ a & b \end{matrix}$$

$$(a, c) = (1, 2) \in R$$

$$\underline{\underline{(a, b)}} \in R$$

$$\underline{\underline{(b, c)}}$$

$$\underline{\underline{R \subset S \times S}}$$

$$\begin{matrix} a & b \\ (3, 5) \end{matrix} \in R, \begin{matrix} c \\ (5, 5) \end{matrix} \in R \Rightarrow \begin{matrix} (a, c) \\ (3, 5) \end{matrix}$$

1. Suppose the relation $R = \{(a, b) \in S \times S \mid |a - b| \text{ is even}\}$ is defined on the set $S = \{3, 4, 5, 6\}$. The relation R on the set S is (are)

(i) Reflexive. $\rightarrow$ If $a \in S \Rightarrow \underline{(a, a) \in R}$. $|a - a| = 0 \xleftarrow{\text{is even}}$

(ii) Symmetric.

(iii) Transitive.

(iv) None of these.

$$\text{If } (a, b) \in R \Rightarrow (b, a) \in R.$$

$$\downarrow \quad \quad \downarrow$$

$$|a - b| \text{ is even.} \quad \rightarrow \quad |b - a| \text{ is even.}$$

$$\text{If } \underline{(a, b) \in R \text{ and } (b, c) \in R} \Rightarrow \underline{(a, c) \in R}.$$

$$\downarrow$$

$$|a - b|$$

$\downarrow$ even

$$\downarrow$$

$$|b - c|$$

$\downarrow$ even.

$$\downarrow$$

$$|a - c| \text{ is even.}$$

$$\swarrow \parallel$$

$$\underline{|a - b + b - c|}$$

$\uparrow$ is even.

$$x > y \Rightarrow \underline{e^x > e^y}$$

$$\text{If } -3 > -4 \Rightarrow \underline{e^{-3} > e^{-4}}$$

$$\underline{100 > 5}$$

$$x = -3 > y = -4$$

$$e^x = 100 > e^y = 5. \quad \underline{e^x > e^y}$$

$$\underline{100x - 4} > \underline{5x - 4}$$

~~$f(x) = xe^x \leftarrow \ln(\ln e)$~~

~~(∞, ∞)~~

~~$x, y \quad f(x) = x^3$~~

2. Which of the following functions $f: \mathbb{R} \rightarrow \mathbb{R}$ are one-one or on-to?

- ☐ $f(x) = x^3 - x \leftarrow$ on-to
☐ $f(x) = x^3 + x \leftarrow$ on-to
☐ $f(x) = 3e^x \leftarrow$ not on-to
☐ $f(x) = x - 1$
- ~~Not one-one~~
one-one
one-one
- both one-one
on-to

~~$$\begin{aligned}
 & \text{If } x > y \\
 & x^3 > y^3 \\
 & \Rightarrow x^3 + x > y^3 + y \\
 & \Rightarrow \underline{\underline{f(x)}} > \underline{\underline{f(y)}}
 \end{aligned}$$~~

$$f(x) = xe^x$$

~~$(0, \infty)$~~

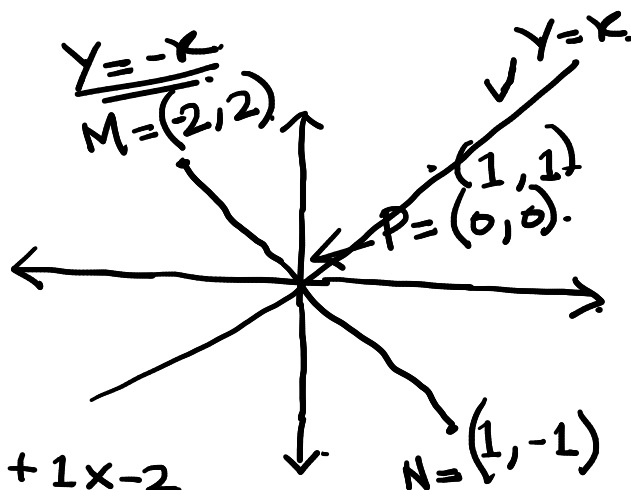
$$f'(x) = e^x + xe^x > \underline{\underline{0}}$$

$$= e^x(x+1)$$

$$\underline{\underline{x < 1}}$$

3. A point P divides the line segment MN such that $MP : PN = 2 : 1$. The coordinates of M and N are $(-2, 2)$ and $(1, -1)$ respectively. What will be the slope of the line passing through P and the point $(1, 1)$?

- ☐ $\frac{4}{3}$
☒ 1
☐ $-\frac{4}{3}$
☐ $\tan(\frac{4}{3})$



x-coordinate

$$= \frac{mx_2 + nx_1}{m+n} = \frac{2 \times 1 + 1 \times -2}{3} = 0$$

y-co

$$= \frac{my_2 + ny_1}{m+n} = 0$$

$P = (0, 0) \rightarrow (1, 1) \rightarrow \textcircled{1}$

$$4y = 3x + 7 \Rightarrow y = \frac{3}{4}x + \frac{7}{4}$$

4. Given the point $(-2, 1)$ and the line $-3x + 4y - 7 = 0$, choose the set of correct options.

☒ The distance between $(-2, 1)$ and the line $-3x + 4y - 7 = 0$ is 2.

☒ The slope of the given line is 5.

☒ The distance between $(-2, 1)$ and the line $-3x + 4y - 7 = 0$ is $\frac{3}{5}$

☒ The y -intercept of the given line is $\frac{7}{4}$

$$4y - 7$$

$$y = x/4$$

$$(m, n)$$

$$(-2, 1)$$

$$-3x + 4y - 7 = 0$$

$$ax + by + c = 0$$

$$\frac{|am + bn + c|}{\sqrt{a^2 + b^2}}$$

$$\left| \frac{6 + 4 - 7}{5} \right| = \frac{3}{5}$$

5. Consider a quadratic function $f(x) = ax^2 + bx + 5$, and $(3, 2)$ is the vertex of f . Then choose the correct options.

- ☐ $f(x) = x^2 - 4x + 5$
☐ $f(x) = \frac{1}{3}x^2 - 2x + 5$
☐ $f(x) = -2x^2 + 5x + 5$
☐ $f(x) = \frac{-2}{3}x^2 + x + 5$

$$f(x) = \underline{ax^2} + bx + 5$$

$$f(3) = \underline{\underline{a \times 3^2}} + 3b + 5$$

$$\begin{aligned} & \swarrow \underline{(3, 2)} \\ & \frac{-b}{2a} \end{aligned}$$

$$\Rightarrow -\frac{b}{2a} = 3$$

$$\Rightarrow 6a + b = 0$$

$$f(3) = 2 \Rightarrow \underline{\underline{9a}} + 3b + 5 = 2$$

$$\Rightarrow 9a + 3b = -3$$

$$\Rightarrow 3a + b = -1 \leftarrow$$

$$3a = 1$$

$$\Rightarrow a = \frac{1}{3}$$

$$b = -6a$$

$$= -6 \times \frac{1}{3} = \underline{\underline{-2}}$$

$$(x-1)$$

$$\underline{\underline{x-2}} = (x-1) - \underline{\underline{1}}$$

6. Consider a polynomial $f(x) = x^4 + 5x^3 + ax + b$, where a and b are constants. The remainder when $f(x)$ is divided by $(x - 2)$ is equal to the remainder when $f(x)$ is divided by $(x + 1)$. Then find value of a .

$$\begin{aligned}
 & f(x) = \underbrace{(x-2)}_0 q(x) + \underbrace{b}_{\text{Constant}} \\
 & f(2) = 16 + 40 + 2a + b = \underline{\underline{56 + 2a + b}} \\
 & f(x) = \underbrace{(x+1)}_0 q'(x) + \underline{b} \quad f(-1) = \underline{\underline{-4 - a + b}} \\
 & \Rightarrow f(2) = f(-1) \\
 & \Rightarrow 56 + 2a + b = -4 - a + b \\
 & \Rightarrow \boxed{a = -20}
 \end{aligned}$$

$$-\frac{16}{9} < 4 < 16$$

7. Solve the following logarithmic equation for x .

$$\log_{10}(x+4) + \log_{10}(x+16) = 1 + 2\log_{10} x$$

$$\underline{\underline{x=4}}$$

$$\log_{10}(x+4) + \log_{10}(x+16) = \log_{10} 10 + \log_{10} x^2$$

$$\log_{10}(x+4)(x+16) = \log_{10} 10x^2$$

$$x^2 + 16x + 4x + 64 = 10x^2$$

$$\Rightarrow x^2 + 20x + 64 = 10x^2 \quad (x+4)(x+16) = 10x^2$$

$$9x^2 - 20x - 64 = 0$$

$$\Rightarrow x = \frac{20 \pm \sqrt{400 + 2304}}{18} = \frac{20 \pm 52}{18}$$

$$\underline{\underline{x=4}}, \quad \frac{-32}{18} = \underline{\underline{\left(-\frac{16}{9}\right)}}$$

$$\ln(2x+3)$$

$$2x+3 > 0 \\ \Rightarrow 2x > -3 \\ \Rightarrow x > -3/2 \quad \log x \quad \text{Ibb } x > 0$$

8. Find the domain of the following functions.

$$a. f(x) = 8 + 5 \ln(2x+3)$$

$$b. f(x) = \ln\left(\frac{x+1}{x}\right)$$

Ibb

$$2x+3 > 0$$

$$\Rightarrow x > -3/2$$

$$(-3/2, \infty)$$

$$\frac{a}{b} = \frac{x+1}{x} > 0$$

$$a/b > 0$$

$$a > 0 \text{ \& } b > 0$$

$$x+1 > 0 \text{ and } x > 0$$

||

$$x > -1 \text{ and } x > 0$$

↓

$$(-1, \infty)$$

$$(0, \infty)$$

$$(0, \infty)$$

$$a < 0 \text{ and } b < 0$$

$$x+1 < 0 \text{ \& } x < 0$$

$$x < -1 \text{ \& } x < 0$$

$$\hookrightarrow (-\infty, -1) \text{ \& } (-\infty, 0)$$

$$(-\infty, -1)$$

Week 1-6
Week-7-9
Week-10-11

Sum. Square Error.

x_i					← $\dots$
y_i					

$W \& T$

$$\underline{Y = mx + c}$$

$$\sum_{i=1}^n (y_i - mx_i - c)^2$$

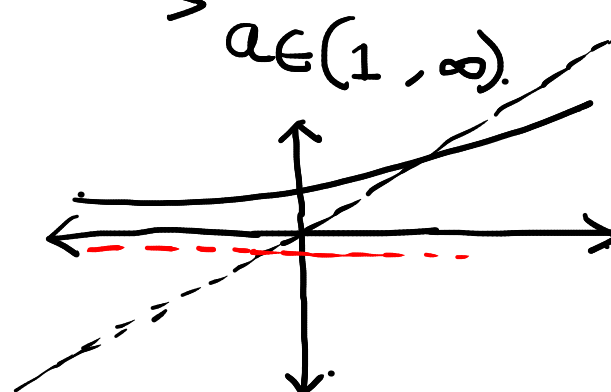
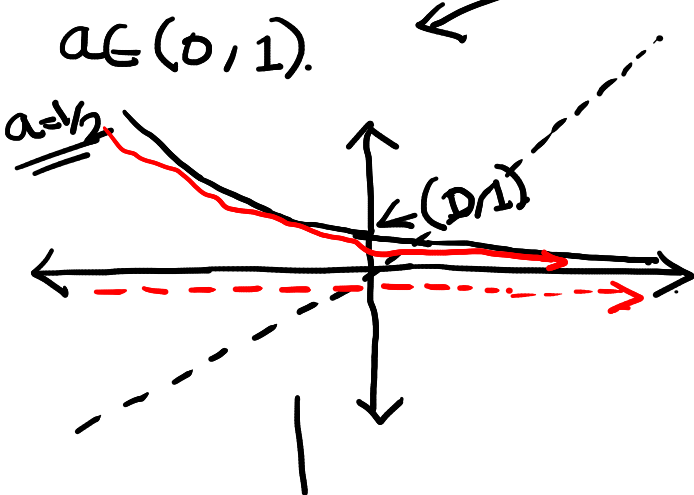
x_i	x_1	1	7	9	$\pi=3$
y_i	y_1	5	2	6	

$$\begin{aligned} \sum_{i=1}^3 (y_i - 3x_i - 2)^2 &= (y_1 - 3x_1 - 2)^2 + \dots + \dots \\ &= (5 - 3 - 2)^2 + (2 - 21 - 2)^2 + (6 - 27 - 2)^2 \\ &= 0 + \boxed{21^2 + 23^2} \end{aligned}$$

$$\underline{Y = 3x + 2}$$

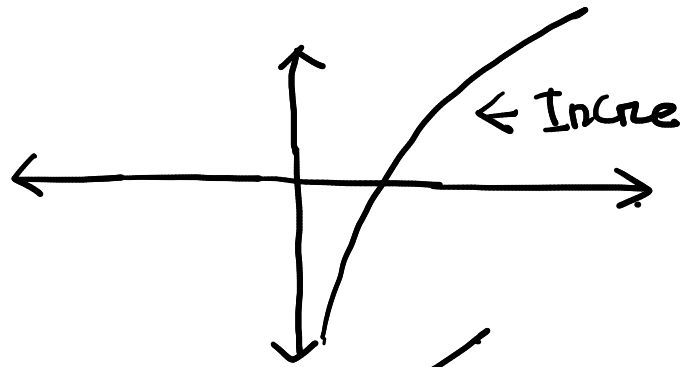
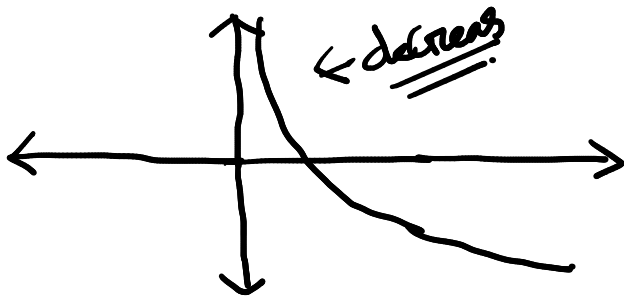
Week-6

$$f(x) = a^x \quad f(x) = \left(\frac{1}{2}\right)^x = \left(\frac{0.5}{a}\right)^x$$



$\log_a$ $a \in (0, 1)$

$\log_a$ $a \in (1, \infty)$



$f(x) = e^{|x|}$
 $f(x) = e^x$

$x^2 |x|$ $\leftarrow$ diff

$$\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{e^{|h|} - 1}{h}$$

$\rightarrow$ L.H.L $|h| = -h$

$$\lim_{h \rightarrow 0} \frac{e^{-h} - 1}{h}$$

R.H.L

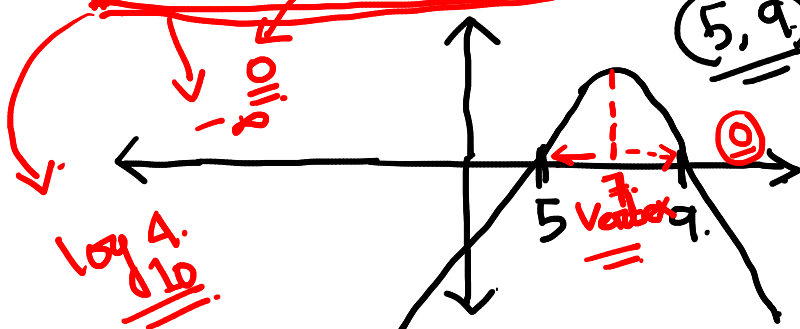
$$\lim_{h \rightarrow 0} \frac{e^h - 1}{h}$$

$\log(-x^2 + 14x - 45)$ $\leftarrow$ Domain
 $(5, 9)$

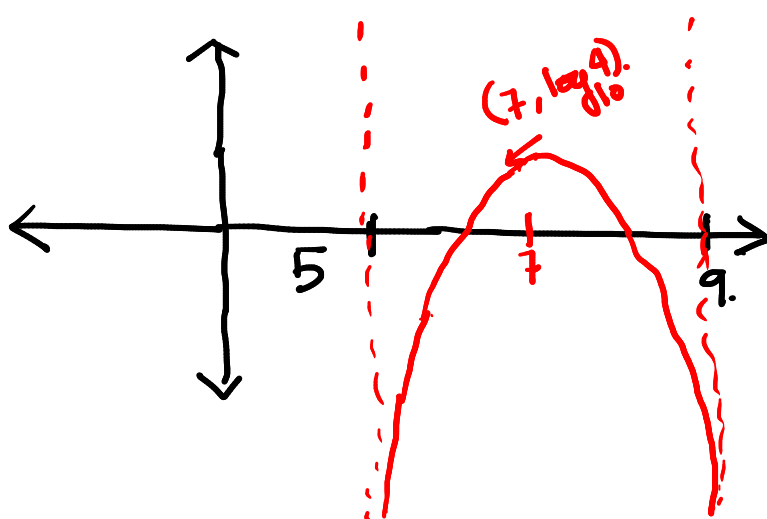
$$-x^2 + 14x - 45 = -(x-9)(x-5)$$

9, 5

$(7, b)$



$\log_b$



$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x) = x^3 + 1$$

$$g: \mathbb{R} \rightarrow \mathbb{R}$$

$$\underline{(f \circ g)(x)} = \underline{(g \circ f)(x)} = x$$

$$f(g(x)) = x$$

$$f(y) = x$$

$$y^3 + 1 = x$$

$$\Rightarrow y^3 = x - 1$$

$$\Rightarrow y = (x - 1)^{1/3}$$

$$\Rightarrow g(x) = (x - 1)^{1/3}$$

$$(g \circ f)(x) = g(x^3 + 1)$$

$\equiv$

$$(x^3 + 1 - 1)^{1/3}$$

$$= \underline{x}$$

$g(x) = y$